Truth values

The truth value of a mathematical statement is not determined in isolation, but always relative to a universe of discourse or context, in which the statement is interpreted. For example, consider the statement

$$\forall x, y \exists z \left((x \neq y) \implies (x < z < y)\right)$$

which may be read as

$$\text{"For any two distinct numbers } x, y \text{, there exists a number } z \text{ strictly between them."}$$

Note that if the statement is interpreted in the context of natural numbers, it is false because there is no natural number between 1 and 2. But if the statement is interpreted in the context of rational numbers, it is true because for any two rational numbers x, y, the number $z = \dfrac{x+y}{2}$ is a rational number between them.

- **Definition 1**
  Let P and Q be mathematical statements interpreted in some context u, the truth value of $P \wedge Q$ in such a context is fully determined by the following truth table:

P	Q	$P \wedge Q$
1	1	1
1	0	0
0	1	0
0	0	0

- **Remark 1**
  The truth table for conjunction can be understood in an even more compact way as follows:
  - **Example 1.1**
    If $\|P\|$ is the truth value of P and $\|Q\|$ is the truth value of Q, then

$$\|P \wedge Q\| = \min\left\{\|P\|, \|Q\|\right\}$$

    This algebraic perspective is useful for replacing structural properties of logical connectives.

- **Definition 2**
  Let P and Q be mathematical statements interpreted in some context u, the truth value of $P \vee Q$ in such a context is fully determined by the following truth table:

P	Q	$P \vee Q$
1	1	1
1	0	1
0	1	1
0	0	0

- **Remark 2**
  The truth table for disjunction can be understood in an even more compact way as follows:
  - **Example 2.1**
    If $\|P\|$ is the truth value of P and $\|Q\|$ is the truth value of Q, then

$$\|P \vee Q\| = \max\left\{\|P\|, \|Q\|\right\}$$

- **Definition 3**
  Let P and Q be mathematical statements interpreted in some context u, the truth value of $P \implies Q$ in such a context is fully determined by the following truth table:

P	Q	$P \implies Q$
1	1	1
1	0	0
0	1	1
0	0	1

- **Remark 3**
  In algebraic form, the truth table can be understood in an even more compact way as follows:
  - **Example 3.1**
    If $\|P\|$ is the truth value of P and $\|Q\|$ is the truth value of Q, then

$$\|P \implies Q\| = \begin{cases} \|Q\| & \text{if } \|P\| = 1 \\ 1 & \text{otherwise} \end{cases}$$

    or in another form:

$$\|P \implies Q\| = \max\left\{1 - \|P\|, \|Q\|\right\}$$

- **Definition 4**
  Let P and Q be mathematical statements interpreted in some context u, the truth value of $P \Leftrightarrow Q$ in such a context is fully determined by the following truth table:

P	Q	$P \Leftrightarrow Q$
1	1	1
1	0	0
0	1	0
0	0	1

- **Remark 4**
  In algebraic form, the truth table can be understood in an even more compact way as follows:
  - **Example 4.1**
    If $\|P\|$ is the truth value of P and $\|Q\|$ is the truth value of Q, then

$$\|P \Leftrightarrow Q\| = \min\left(1 - \|P\| + \|Q\|, 1 - \|Q\| + \|P\|\right)$$

- **Definition 5**
  Let P be a mathematical statement interpreted in some context u, the truth value of $\neg P$ in such a context is fully determined by the following truth table:

P	$\neg P$
1	0
0	1

- **Remark 5**
  In algebraic form, the truth table can be understood in an even more compact way as follows:
  - **Example 5.1**
    If $\|P\|$ is the truth value of P, then

$$\|\neg P\| = 1 - \|P\|$$

We now turn to the question of how truth values are assigned to statements involving quantifiers.

- **Definition 6**
  A universally quantified statement of the form $\forall x P(x)$ is true if and only if $P(x)$ is true for all possible values of x. Note that the universally quantified statement $\forall x P(x)$ is false if $P(x)$ is false for some value of x.
  An existentially quantified statement of the form $\exists x P(x)$ is true if $P(x)$ is true for some value of x. Note that the existentially quantified statement $\exists x P(x)$ is false if $P(x)$ is false for all possible values of x.
- **Remark 6**
  In algebraic form, the truth values of the qu/antifiers $\forall x P(x)$ and $\exists x P(x)$ in a context u are:
  - **Example 6.1**

$$\|\forall x P(x)\| = \min_{x \text{ takes a value in } u} \{\|P(x)\|\}$$

$$\|\exists x P(x)\| = \max_{x \text{ takes a value in } u} \{\|P(x)\|\}$$

- **Definition 7**
  We say that two mathematical statements P and Q are logically equivalent, denoted by $P \equiv Q$, if for any context u we obtain $\|P\| = \|Q\|$.
  - **Example 7.1**
    Let P and Q be two mathematical statements, then $\neg(P \wedge Q) \equiv (\neg P \vee \neg Q)$.

P	Q	$P \wedge Q$	$\neg(P \wedge Q)$	$\neg P$	$\neg Q$	$(\neg P \vee \neg Q)$
1	1	1	0	0	0	0
1	0	0	1	0	1	1
0	1	0	1	1	0	1
0	0	0	1	1	1	1

    Therefore, $\neg(P \wedge Q) \equiv (\neg P \vee \neg Q)$.
  - **Example 7.2**
    Let P and Q be two mathematical statements, then $\neg(P \implies Q) \equiv P \wedge \neg Q$.

P	Q	$P \implies Q$	$\neg(P \implies Q)$	$\neg Q$	$(P \wedge \neg Q)$
1	1	1	0	0	0
1	0	0	1	1	1
0	1	1	0	0	0
0	0	1	0	1	0

    Therefore, $\neg(P \implies Q) \equiv P \wedge \neg Q$.
  - **Example 7.3**
    Determine if $\forall x P(x) \equiv \neg \exists x (\neg P(x))$.
    - **Solution 7.3.1**
      We consider two possible cases for the truth value of $\forall x P(x)$ in any given context u:

      Case 1: If $\forall x P(x)$ is true, then $\|\forall x P(x)\| = 1$. This implies that $\|P(x)\| = 1$ for all x in u. Consequently, the truth value of its negation is $\|\neg P(x)\| = 1 - \|P(x)\| = 0$ for all x in u. Taking the maximum over all x, we get:

$$\|\exists x(\neg P(x))\| = \max_x \{\|\neg P(x)\|\} = 0$$

Thus, its negation evaluates to:

$$\|\neg \exists x\,(\neg P(x))\| = 1 - 0 = 1$$

Case 2: If $\forall x P(x)$ is false, then $\|\forall x P(x)\| = 0$. This implies that there exists at least one element x_0 in u such that $\|P(x_0)\| = 0$. For this specific element, we have $\|\neg P(x_0)\| = 1 - \|P(x_0)\| = 1$. Since there is at least one element where $\neg P(x)$ is true, taking the maximum yields:

$$\|\exists x(\neg P(x))\| = \max_x \{\|\neg P(x)\|\} = 1$$

Thus, its negation evaluates to:

$$\|\neg \exists x\,(\neg P(x))\| = 1 - 1 = 0$$

In both cases, we obtain:

$$\|\forall x P(x)\| = \|\neg \exists x\,(\neg P(x))\|$$

By Definition 2.7, we conclude that $\forall x P(x) \equiv \neg \exists x\,(\neg P(x))$.

- **Example 7.4**
  Let P, Q and R be three mathematical statements, we want to prove that the propositions

$$\begin{aligned} &A : P \implies (Q \implies R) \\ &B : (P \wedge Q) \implies R \end{aligned}$$

  are equivalent.

  - **Solution 7.4.1**
    We have

P	Q	R	$Q \implies R$	$P \wedge Q$	$P \implies (Q \implies R)$	$(P \wedge Q) \implies R$
1	1	1	1	1	1	1
1	1	0	0	1	0	0
1	0	1	1	0	1	1
1	0	0	1	0	1	1
0	1	1	1	0	1	1
0	1	0	0	0	1	1
0	0	1	1	0	1	1
0	0	0	1	0	1	1

  Therefore, $A \equiv B$.

- **Example 7.5**
  Let P, Q and R be three mathematical statements, we want to prove that the propositions

$$\begin{aligned} &A : (P \implies Q) \wedge (Q \implies R) \\ &B : (\neg P \vee Q) \wedge (Q \vee R) \end{aligned}$$

  are equivalent.

  - **Solution 7.5.1**
    Case 1: A is false if and only if

$$(P \implies Q) \text{ is false or } (\neg Q \implies R) \text{ is false}$$

    which means

$$\begin{pmatrix} P \text{ is true} \\ \text{and} \\ Q \text{ is false} \end{pmatrix} \text{ or } \begin{pmatrix} Q \text{ is false} \\ \text{and} \\ R \text{ is false} \end{pmatrix}$$

Case 2: B is false if and only if

$$(\neg P \vee Q) \text{ is false or } (Q \vee R) \text{ is false}$$

which means

$$\begin{pmatrix} P \text{ is true} \\ \text{and} \\ Q \text{ is false} \end{pmatrix} \text{ or } \begin{pmatrix} Q \text{ is false} \\ \text{and} \\ R \text{ is false} \end{pmatrix}$$

In conclusion, both propositions are false exactly in the same cases.
Therefore, $A \equiv B$.

- **Example 7.6**
  Let P, Q and R be three mathematical statements, we want to prove that the propositions

$$\begin{aligned} A &: P \implies (Q \implies R) \\ B &: (P \implies Q) \implies R \end{aligned}$$

  are not equivalent.

  - **Solution 7.6.1**
    if P is false, Q is true and R is false, then the proposition A is true and the proposition B is false, then A, B are not equivalent.

- **Example 7.7**
  Are $A : \neg(P \wedge Q), B : \neg P \wedge \neg Q$ are equivalent?

  - **Solution 7.7.1**
    If P is true and Q is false, then A is true and B is false, then A, B are not equivalent.

- **Example 7.8**
  Are $A : (P \implies Q) \wedge R, B : P \implies (Q \wedge R)$ are equivalent?

  - **Solution 7.8.1**
    If P is false, then B is always true. In this case, if R is false, A is false, then A, B are not equivalent.